HEX [PI] 7.0

The ten-year replay: what was measured, and how to run and watch the system

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The result in five lines

HEX [PI] 7.0 was run once through history, making every call before its outcome and starting every series from nothing: 280,252 calls on 43 series.

On next-day **move size** it gave what happened more probability than the series' own record did, by a margin clear of zero, on the 17 markets it was developed on *and* on ten currencies it had never seen. On **temperature** it did the same for direction at three stations it had never seen.

On next-day **market direction** it has **no verified edge**, and the product says so.

Learning itself adds value: the engine left to learn beats the same engine frozen after its first 500 calls.

The code that was tested is the code that is deployed: the two agree to 10^{-13}.

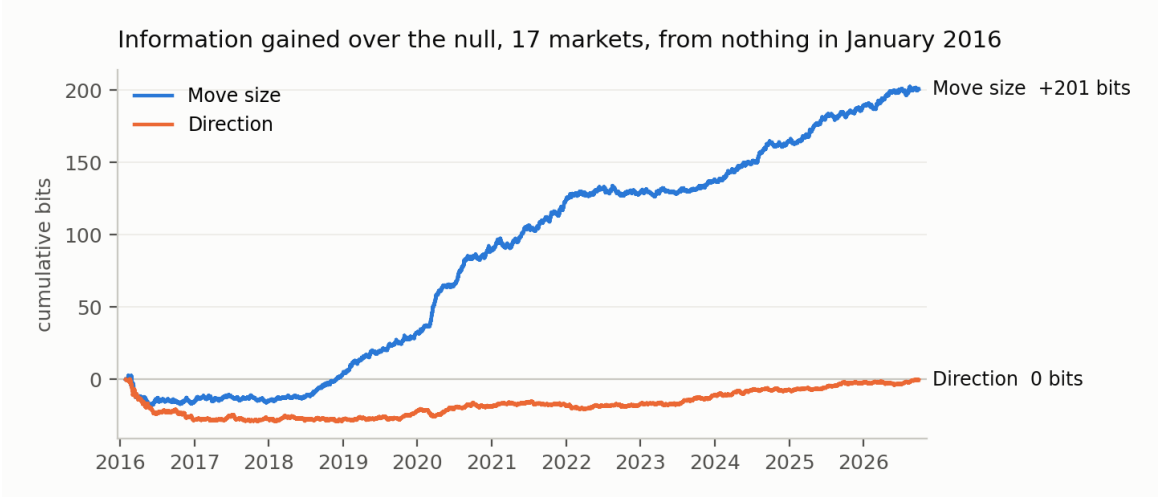
1 The measure

A call is scored by its **information gain** over the null: $g = \ln p(y) - \ln q_0(y)$, where p is HEX's probability, q_0 is what the series' own record already implied, and y is what happened. It is reported in bits per call. Positive means HEX gave what happened more probability than the obvious did. Beside it: the share of calls right, and "expected", the share right if the null alone had made every call. Intervals are 95%, from a block bootstrap over 20-day blocks so that calls on neighbouring days are not treated as independent. A read is **verified** when the interval is above zero.

2 Headline results

	Calls	HEX right	Expected	Bits per call over the null [95%]	Status
Markets (17, 2016–2026), move size	46,478	57.9%	57.2%	+0.0043 [+0.0030, +0.0056]	verified
Markets (17, 2016–2026), direction	46,478	51.4%	51.2%	0.0000 [−0.0007, +0.0006]	no verified edge
New: 10 currencies, move size	26,540	62.1%	61.2%	+0.0082 [+0.0060, +0.0106]	verified
New: 10 currencies, direction	26,540	54.2%	54.0%	+0.0011 [−0.0002, +0.0025]	no verified edge
New: 3 temperature stations, move size	29,408	63.1%	61.6%	+0.0186 [+0.0158, +0.0212]	verified
New: 3 temperature stations, direction	29,408	61.0%	54.6%	+0.0466 [+0.0430, +0.0499]	verified
Chicago temperature, move size	24,678	55.1%	54.1%	+0.0048 [+0.0035, +0.0061]	verified
Chicago temperature, direction	24,678	62.9%	50.7%	+0.0644 [+0.0598, +0.0689]	verified
Racing (12 series), move size	13,022	52.9%	49.7%	+0.0123 [+0.0001, +0.0295]	verified
Racing (12 series), direction	13,022	74.8%	75.0%	−0.0017 [−0.0020, −0.0014]	no verified edge

New marks series never used in any earlier experiment. The rule for including them was written before any run.



The curve can go down as well as up and cannot hide a bad period. Move size loses a little in 2016, while the engine has seen almost nothing, is flat to 2018, and then gains steadily. Direction ends where it started.

3 How the replay was run

1. **One pass, in order.** For each series and each timestamp: reveal the outcome, score, learn, predict, commit. No fitting on later data, no recalculating an earlier call.
2. **From nothing.** Every series starts with no history. The 17 markets start on 1 January 2016.
3. **A specification frozen first.** The engine, the data rules and seven predictions were written down before the run (`validation/prereg_replay.md`).
4. **A ledger of every call** (`validation/replay/hex7_ledger.parquet`): the probabilities of the null, 7.0, the previous engine 6.5 and a frozen copy; the outcome; and hashes of the engine's state, of the code and of the data seen so far. The learned coefficients are kept every 25th call, and the final state of every series is kept too.

Three checks

1. **No call saw its own future.** The engine was rerun on truncated histories at 258 sampled points and compared with the ledger: largest difference 0.
2. **The product code is the tested code.** On identical inputs the TypeScript engines and their Python references agree to 10^{-13} (`tests/parity.mjs`).
3. **The product's nightly path gives the same calls.** The nightly entry point was run on only what was known on 680 sampled nights: 2,720 calls, none different. And the product code's 92,888 calls on the 17 markets, computed from prices independently of the reference, match the replay's to 10^{-14}.

What is development and what is new

Read the 17-market result as a development backtest

The 17 market series, the Chicago temperature series and the racing series were used while designing every version of the engine. A result on them shows how the final engine would have behaved; it is not an untouched test.

The untouched tests are the ten currencies and the three temperature stations. Currencies: every country in the exchange-rate file other than the twelve already used, kept if it had at least 2,000 observations and a change on at least half its days. All ten passed: Brazil (99%), China (94%), Denmark (100%), Hong Kong (94%), Malaysia (93%), Singapore (99%), South Korea (99%), Taiwan (92%), Thailand (94%), Venezuela (70%) (share of days with a change). The list therefore includes a currency in a pegged band

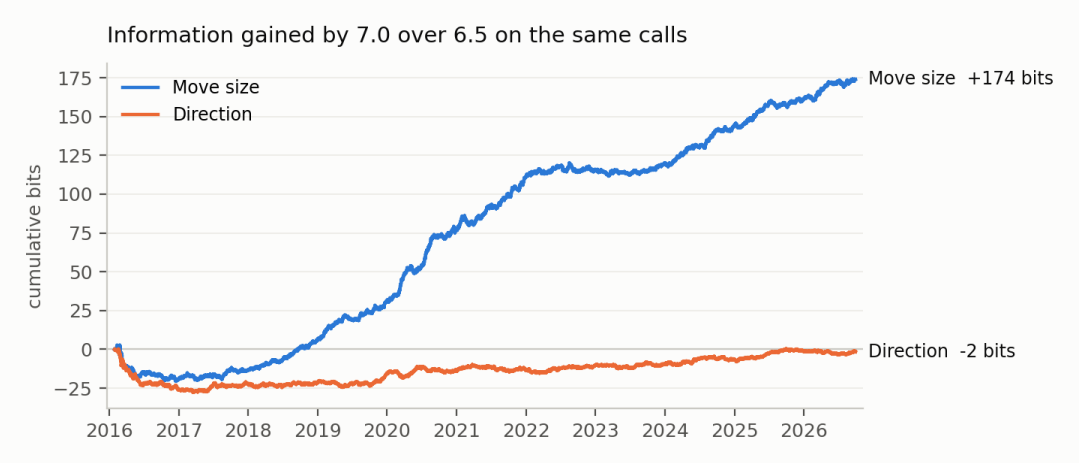
and one in crisis; they were kept because the rule written beforehand kept them. Temperature: every public daily series reachable with at least 1,000 days, daily maximum, one per station: Melbourne 1981–1990, Seattle 1948–2017, New York 2012–2015.

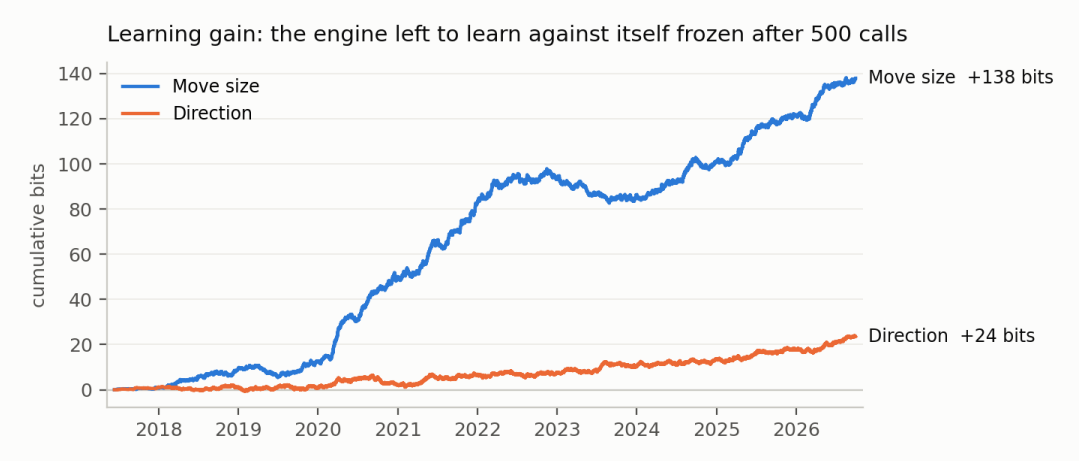
The third test is live: calls made from the release of 7.0, which the ledger collects from the first night.

4 Full results

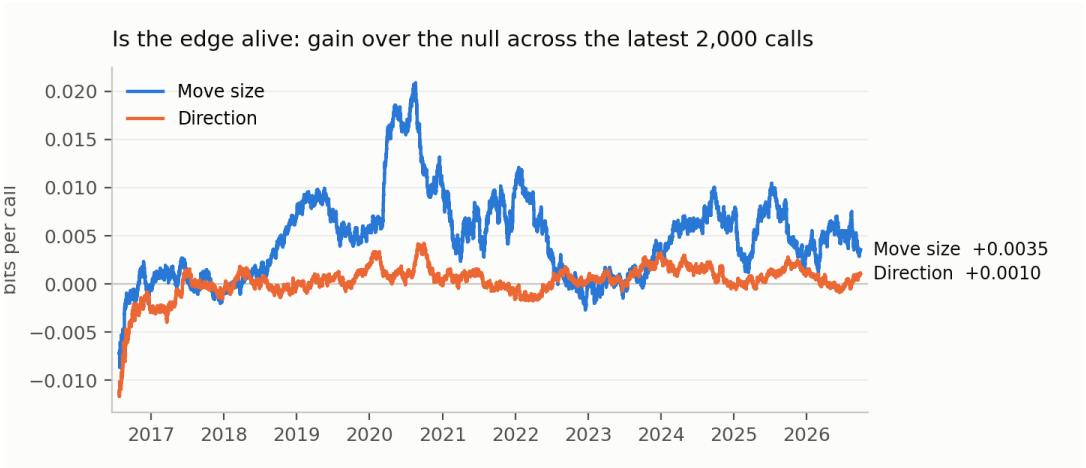
	6.5 right	Over 6.5 [95%]	Learning gain [95%]	Macro	Brier skill	Cal. error	Top tenth
Markets (17, 2016–2026), move size	57.4%	+0.0037 [+0.0025, +0.0050]	+0.0036 [+0.0025, +0.0048]	+0.0042	0.62%	0.013	74 / 70
Markets (17, 2016–2026), direction	51.3%	0.0000 [−0.0007, +0.0006]	+0.0006 [+0.0001, +0.0011]	−0.0001	0.01%	0.016	60 / 56
New: 10 currencies, move size	61.5%	+0.0035 [+0.0018, +0.0053]	+0.0096 [+0.0071, +0.0121]	+0.0082	1.15%	0.009	90 / 89
New: 10 currencies, direction	54.1%	+0.0001 [−0.0013, +0.0014]	+0.0022 [+0.0008, +0.0035]	+0.0011	0.15%	0.017	80 / 78
New: 3 temperature stations, move size	62.1%	+0.0130 [+0.0109, +0.0152]	+0.0084 [+0.0066, +0.0103]	+0.0245	2.68%	0.023	78 / 72
New: 3 temperature stations, direction	57.0%	+0.0310 [+0.0278, +0.0342]	+0.0068 [+0.0054, +0.0082]	+0.0607	6.29%	0.009	74 / 77
Chicago temperature, move size	54.5%	+0.0038 [+0.0027, +0.0049]	+0.0042 [+0.0030, +0.0053]	+0.0048	0.65%	0.018	68 / 63
Chicago temperature, direction	57.6%	+0.0417 [+0.0375, +0.0459]	+0.0065 [+0.0049, +0.0082]	+0.0644	8.67%	0.009	75 / 75
Racing (12 series), move size	51.6%	+0.0068 [+0.0005, +0.0141]	+0.0014 [−0.0003, +0.0038]	+0.0091	1.65%	0.023	69 / 66
Racing (12 series), direction	74.9%	−0.0008 [−0.0015, −0.0003]	+0.0004 [−0.0005, +0.0017]	−0.0018	-0.25%	0.014	99 / 100

Over 6.5: bits per call gained over the previous engine on the same calls. *Learning gain*: the engine left to learn against the same engine with its coefficients and influences frozen after its first 500 calls on a series (the null keeps moving in both). *Macro*: bits per call over the null averaged over series, so that a long series does not outweigh a short one. *Brier skill*: reduction in squared error against the null. *Cal. error*: mean gap between stated probability and observed frequency over ten bins. *Top tenth*: stated strength / share right, in percent, on the strongest tenth of calls.



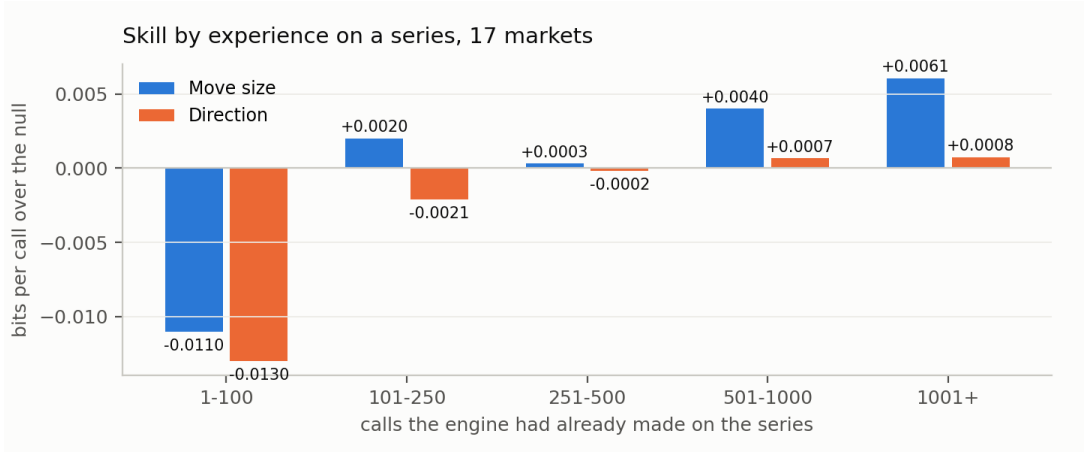


Is the edge alive now?



Gain over the null across the latest 2,000 calls. For move size it is above zero for most of the period from 2019; it dips to zero in 2022–2023 and recovers.

Does it get better with experience?



Calls already made	Calls	Move size Bits/call	Brier skill	Right − expected	Cal. error	Direction Bits/call	Cal. error
1–100	1,700	−0.0110	-1.17%	−0.06	0.062	−0.0130	0.081
101–250	2,550	+0.0020	0.29%	+0.20	0.043	−0.0021	0.037
251–500	4,250	+0.0003	0.06%	+0.00	0.014	−0.0002	0.012
501–1000	8,500	+0.0040	0.58%	+0.42	0.020	+0.0007	0.018
1001+	29,478	+0.0061	0.85%	+0.94	0.015	+0.0008	0.010

Aligning by experience instead of by calendar separates "the world became easier" from "the engine became more experienced". The first hundred calls on a new series cost information; the engine is level with the null by 500 and ahead after. Its stated probabilities also become more accurate: calibration error falls from 0.06 to 0.015.

By year, 17 markets

	Move size			Direction		
Year	Over null	Over 6.5	Right / expected	Over null	Over 6.5	Right / expected
2016	−0.0040	−0.0049	53.2% / 52.9%	−0.0071	−0.0066	50.4% / 51.2%
2017	+0.0002	+0.0014	60.2% / 60.4%	+0.0001	+0.0008	52.2% / 52.0%
2018	+0.0044	+0.0046	55.3% / 55.3%	0.0000	+0.0003	50.7% / 50.1%
2019	+0.0065	+0.0058	59.5% / 58.2%	+0.0014	+0.0014	52.6% / 51.1%
2020	+0.0128	+0.0102	57.4% / 55.6%	+0.0006	+0.0003	51.8% / 51.5%
2021	+0.0082	+0.0081	56.5% / 54.4%	+0.0001	+0.0001	51.4% / 51.8%
2022	+0.0011	+0.0008	60.3% / 60.4%	+0.0003	+0.0008	51.0% / 51.4%
2023	+0.0017	+0.0009	57.2% / 57.0%	+0.0015	+0.0003	52.0% / 51.4%
2024	+0.0063	+0.0058	60.5% / 59.5%	+0.0007	+0.0005	51.0% / 51.3%
2025	+0.0056	+0.0040	57.9% / 57.2%	+0.0013	+0.0013	50.5% / 49.9%
2026	+0.0035	+0.0037	58.0% / 57.7%	+0.0004	−0.0002	51.7% / 51.3%

By series

Bits per call over the null, and right / expected.

		Move size		Direction	
Series	Calls	Bits/call	Right / expected	Bits/call	Right / expected
Markets, development					
BRENT	2,698	+0.0062	59.0% / 57.4%	−0.0007	52.1% / 52.3%
BTC	3,898	+0.0062	60.4% / 59.8%	+0.0010	52.9% / 51.1%
Australia	2,654	−0.0011	56.0% / 56.3%	−0.0013	50.8% / 50.9%
Canada	2,654	−0.0009	59.5% / 59.7%	−0.0011	49.2% / 48.1%
Euro	2,654	−0.0006	56.6% / 56.6%	−0.0011	49.0% / 49.7%
India	2,654	+0.0036	60.6% / 60.1%	+0.0002	51.3% / 50.3%
Japan	2,654	+0.0035	60.2% / 59.2%	−0.0012	50.9% / 51.1%
Mexico	2,654	+0.0150	59.0% / 57.0%	−0.0010	52.9% / 52.8%
NewZealand	2,654	−0.0011	57.5% / 57.5%	−0.0013	48.8% / 49.0%
Norway	2,654	+0.0004	55.6% / 55.4%	−0.0011	49.8% / 49.8%
SouthAfrica	2,654	+0.0016	54.6% / 54.8%	−0.0004	51.7% / 51.7%
Sweden	2,654	+0.0001	54.0% / 54.4%	−0.0013	49.1% / 49.5%
Switzerland	2,654	+0.0011	54.2% / 53.9%	−0.0010	51.6% / 51.2%
UnitedKingdom	2,654	−0.0009	57.8% / 58.1%	−0.0011	48.2% / 49.4%
NATGAS	2,676	+0.0007	60.1% / 59.5%	+0.0008	55.4% / 55.4%
VIX	2,700	+0.0318	58.6% / 54.2%	+0.0108	56.5% / 54.8%
WTI	2,658	+0.0065	58.8% / 57.0%	−0.0010	52.3% / 53.0%
Currencies never used before					
Brazil	2,654	+0.0001	59.3% / 58.8%	−0.0012	49.3% / 50.9%
China	2,654	+0.0159	61.2% / 59.8%	+0.0036	53.7% / 52.2%
Denmark	2,654	−0.0009	56.6% / 56.2%	−0.0011	50.8% / 51.3%
HongKong	2,654	+0.0488	69.5% / 65.6%	+0.0006	51.0% / 51.5%
Malaysia	2,654	+0.0044	60.1% / 59.0%	0.0000	51.6% / 51.4%
Singapore	2,654	+0.0011	57.2% / 56.5%	0.0000	50.3% / 51.2%
SouthKorea	2,654	−0.0008	54.5% / 54.8%	−0.0011	50.0% / 50.7%
Taiwan	2,654	+0.0036	57.4% / 57.0%	+0.0021	53.1% / 50.7%
Thailand	2,654	+0.0010	59.6% / 59.5%	+0.0014	53.8% / 53.0%
Venezuela	2,654	+0.0086	85.6% / 84.6%	+0.0069	78.3% / 77.4%
Temperature never used before					
Melbourne	3,622	+0.0503	64.8% / 59.0%	+0.0651	63.1% / 54.0%
NewYork	1,433	+0.0088	58.0% / 58.5%	+0.0749	64.5% / 45.4%
Seattle	24,353	+0.0144	63.2% / 62.2%	+0.0421	60.5% / 55.2%

5 Predictions written before the run

	Prediction	Outcome
1	Market move size: 0.002 to 0.006 bits per call, clear of zero	Held: +0.0043 [+0.0030, +0.0056]
2	Market direction: under 0.0015	Held: 0.0000 [−0.0007, +0.0006]
3	7.0 over 6.5 on market move size: positive, clear of zero	Held: +0.0037 [+0.0025, +0.0050]
4	Learning gain positive on move size; within 0.001 of zero on direction	Held: +0.0036 and +0.0006
5	Unused currencies: move size positive on the macro-average, direction not clear of zero	Held
6	Temperature replication: direction gain positive and at least 58% right at every station	Held: 63.1%, 64.5%, 60.5%
7	The strength map does not lower information gain on move size	Neither way: −0.0001 [−0.0010, +0.0007]

One amendment after the first run. Comparing the product code's calls with the reference's showed differences on currencies quoted to four decimals. When a price returned exactly to an earlier price, the rank of that tie was being decided by the last binary digit of a sum of log changes. Levels are now rounded to nine decimals before they are ranked, in the engine and in the null. Everything was rerun; no prediction was changed. The racing null improved as a side effect (it has many exact ties), which is why racing direction shows HEX slightly behind the null.

6 The product's own loop, run through the same ten years

The product's code and its ledger functions were run night by night on the 17 markets with the database in memory (`validation/timesim`): score, state the calls with that night's strength maps, write them to the ledger, then the nightly update. It began exactly as the product is configured to begin, showing 7.0 for both reads, but with an engine that had seen nothing.

What the rules did: 2016-04-04, direction: 6.5 switched in; 2016-05-22, move size: 6.5 switched in; 2017-07-03, move size: not verified; 2019-05-20, move size: 7.0 switched in; 2020-03-25, move size: verified.

So from a cold start the system corrected itself: it took 7.0 down within months while it was still behind, switched move size back to 7.0 when the record was decisive, and verified it in 2020. It never switched direction back, and direction ended not verified. In the product the engine is not cold: each series is walked from the start of its history every night, so its calls are those of an engine with thousands of calls of experience.

Final status rows: move size shows 7.0, verified, +0.00421 [+0.00306, +0.00535] bits per call over the null, right 57.9% against 57.4% for 6.5. Direction shows 6.5, not verified.

7 What can and cannot be said

The claim, worded as it should be

"HEX [PI] 7.0 demonstrated positive information gain over its pre-specified baseline on next-day move size, under one-pass walk-forward evaluation, on 17 market series from 2016 to 2026 and on ten currency series it had never been tested on; and on next-day direction of daily maximum temperature at three stations it had never been tested on. It has no verified edge on next-day market direction."

- Do not say "proven accuracy". Finite history cannot prove the future.
- The edge on market move size is small: about half a percentage point of calls right, 0.004 bits per call. It is real on this evidence and it is not large.
- The temperature replication is three stations. The currency replication is ten series from one source.
- Racing direction is slightly *behind* the null. The null is already right three times in four there.
- Text, letter-number and ontology layers are not part of 7.0 and nothing here measures them.

8 Running the system

What runs, and when

Job	Trigger	What it does
Nightly update	daily, 10:00 UTC	`calibrateCognition`: the learning loop, then status
Signal post	three times a day	`discordMorningSignal`: reads the tracked list, posts to Discord
X engagement	every 30 minutes	`xEngagement`
Chat	a user message	`think`: reads a series when needed; logs the call
On demand	a signed-in user	`walkForwardAccuracy`: the one-pass test on live data, with timings

State

The engine stores no state between runs. Each time it is asked, it walks the series from its first bar. There is no checkpoint to corrupt, migrate or roll back; the deployed engine and the replay are the same computation; a code change takes effect for every series on the next run.

Two entities are written: `CallLedger` (one row per series, read, version and day, plus calls made in chat) and `EngineStatus` (one row per read: the version shown, the verified flag, the running tally, the strength maps, the precision metrics).

Cost and scaling

One engine run for one series (both versions, both reads; 2,700 to 3,900 daily bars; two-core container)	150 to 280 ms, mean 170 ms
One nightly update, 7 tracked series	about 1.5 s of engine time, plus seven data fetches
One nightly update at the cap of 27 series	about 6 s of engine time
One chat read of a ticker	one engine run
Ledger growth, 7 series (28 rows a night, about 1.5 kB each)	about 15 MB a year
Ledger growth at the cap of 27 series	about 60 MB a year

Time is linear in the length of a series (capped at 5,475 days) and in the number of series. If engine time ever matters (many more series, or intraday bars), the walk can be made incremental by saving each series' state after its last bar; the state is small. That would give up the "no stored state" property and would need its own parity test. Nothing today requires it.

What to watch

The nightly update returns `ok`, `series`, `failed`, `newly_scored`, `newly_logged`, `elapsed_ms`, `per_series_ms` and a `status` for each read.

Watch	Healthy	If not
`ok`	true	the job threw; see the function log
`failed`	empty	a ticker's data could not be fetched; it is skipped tonight
`newly_logged`	4 × series	fewer: a rerun, or failures
`newly_scored`	about 4 × series	zero for several nights: bars are not arriving
`elapsed_ms`, `per_series_ms`	seconds	growth here is the scaling signal
`live_engine`, `decision`	`hex-7.0`, `hold`	`demote`: 6.5 is clearly ahead on the live record; read `reason`
`verified`	move size true, direction false	a change means the live record became decisive
`bits_candidate` and its interval	positive for move size over time	the live measure of whether the edge is alive
`hit_candidate` beside `hit_baseline`	at or above	share right against what the null alone gets

Two limits: the nightly update reads the 5,000 most recently scored rows (46 nights at the cap of 27 series), so an outage longer than that leaves those days out of the tally (they are never double counted); and a day enters the tally five days after its calls.

Before pressing Publish on Base44

1. `CallLedger` and `EngineStatus` exist with the fields in `base44/entities/`. New in 7.0 — `CallLedger`: `log_score`, `info_gain`, `learned`, `part_structure`. `EngineStatus`: `verified`, the six `bits_*` fields, `closeness_candidate`, `closeness_control`, `calibration_control`, `tally`. The engine names are now `state-6.5` and `hex-7.0`.
2. The nightly workflow is active.
3. Run `calibrateCognition` once by hand and read its response against the table above.
4. Sign-in still works.

Nothing in the ledger needs migrating: no earlier version has run live.

Reproducing every number

```
python3 validation/replay.py          # the one-pass replay (about 4 minutes)
python3 validation/replay_report.py   # every number and chart, from the ledger
node tests/parity.mjs                 # product code against the reference
```

The data are public and are listed in `validation/README.md`.

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